

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO2:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

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QUESTIONS: 10

Total Marks: 50

Annexure A
Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality
2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations
3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.
4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance.
5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

ANSWERS:

1. Ride-Sharing Company Using Reinforcement Learning

Solution

Introduction

A ride-sharing company can use Reinforcement Learning (RL) to efficiently match drivers with passengers. The objective is to minimize passenger waiting time, maximize driver utilization, and improve overall service quality.

RL Framework

Component	Description
Agent	Ride-sharing matching system
Environment	City road network with drivers and passengers
State Space	Driver location, passenger location, traffic conditions, driver availability, estimated travel time
Action Space	Assign a driver to a passenger, reassign a driver, delay assignment, reject request
Reward Function	+50 for successful ride completion, +20 for reduced waiting time, +10 for high customer rating, -20 for long waiting time, -30 for cancelled rides
Policy	Select the driver that maximizes long-term reward by minimizing waiting time and travel distance

Learning Process

1. Observe the current state.
2. Select the best driver assignment.
3. Execute the assignment.
4. Receive a reward based on ride completion and customer satisfaction.
5. Update the policy for future decisions.

Conclusion

The RL framework enables dynamic driver-passenger matching, reducing waiting time, improving service quality, and increasing operational efficiency.

2. E-Learning Platform Using Exploration and Exploitation

Solution

Introduction

An e-learning platform uses Reinforcement Learning to recommend personalized study materials based on each student's learning behavior.

Exploration vs. Exploitation

Exploration

- Recommends new or less-used learning materials.
- Helps discover resources that may improve learning.
- Prevents repetitive recommendations.

Exploitation

- Recommends study materials that have previously helped the student.
- Improves learning efficiency using known successful resources.

RL Components

Component	Description
Agent	Recommendation system
Environment	Student learning platform
States	Student progress, quiz scores, learning history
Actions	Recommend video, notes, quiz, or practice problems
Rewards	+20 for quiz improvement, +10 for lesson completion, -10 for skipped content

Benefits

- Personalized learning experience.
- Better engagement.
- Improved academic performance.
- Continuous adaptation to changing learning needs.

Conclusion

Balancing exploration and exploitation helps the system recommend both familiar and new learning resources, leading to better long-term learning outcomes.

3. Robotic Vacuum Cleaner Using Reinforcement Learning

Solution

Introduction

A robotic vacuum cleaner uses Reinforcement Learning to clean rooms efficiently while avoiding obstacles and minimizing cleaning time.

RL Framework

Component	Description
Agent	Vacuum cleaning robot
Environment	House or office floor
States	Robot position, cleaned areas, battery level, obstacle locations
Actions	Move forward, backward, left, right, stop, return to charging station
Rewards	+10 for cleaning a new area, +50 for completing cleaning, -20 for collisions, -10 for revisiting cleaned areas

Policy

The policy determines the best action in each state to maximize cleaning efficiency while avoiding obstacles and conserving battery power.

Value Function

The value function estimates the expected future reward of each state. As the robot gains experience, it learns which paths lead to faster and more complete cleaning, improving its overall performance over time.

Conclusion

Using policies and value functions allows the robotic vacuum cleaner to continuously improve cleaning efficiency and obstacle avoidance.

4. Cryptocurrency Trading System Using Reinforcement Learning

Solution

Introduction

A cryptocurrency trading system uses Reinforcement Learning to decide whether to buy, sell, or hold assets based on market conditions.

RL Components

Component	Description
Agent	Trading system
Environment	Cryptocurrency market
States	Current price, trading volume, market trends, portfolio value
Actions	Buy, Sell, Hold
Rewards	Profit earned or loss incurred after each action

Challenges in Reward Design

- Markets are highly volatile.
- Short-term profits may conflict with long-term gains.
- Delayed rewards make it difficult to evaluate decisions immediately.
- Transaction fees and risks must be considered.

Effects of Poor Reward Design

- Encourages excessive buying and selling.
- Increases financial losses.
- Promotes risky investment strategies.
- May ignore long-term profitability.
- Results in unstable learning and poor decision-making.

Conclusion

A well-designed reward function should balance profit, risk, and long-term investment goals to ensure stable and effective trading performance.

5. Smart Grid System Using Reinforcement Learning

Solution

Introduction

A smart grid uses Reinforcement Learning to manage electricity distribution efficiently while balancing supply, demand, and energy consumption.

RL Model

Component	Description
Agent	Smart grid controller
Environment	Electricity distribution network
State Space	Electricity demand, power generation, battery storage, weather conditions, grid load
Action Space	Increase power supply, decrease supply, store energy, release stored energy, switch energy sources
Reward Mechanism	+50 for balanced supply and demand, +20 for efficient energy usage, -30 for power outages, -20 for energy wastage

Working

1. Monitor electricity demand.
2. Observe the current grid state.
3. Select the appropriate action.
4. Distribute electricity.
5. Receive a reward based on efficiency and reliability.
6. Update the policy for future decisions.

Benefits

- Reduces power wastage.
- Improves energy efficiency.
- Prevents overloads and outages.
- Supports renewable energy integration.
- Continuously adapts to changing demand.

Conclusion

Reinforcement Learning enables smart grids to make intelligent, real-time electricity distribution decisions, improving efficiency, reliability, and sustainability.

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand